

Unsupervised GAN-Based Intrusion Detection System Using Temporal Convolutional Networks and Self-Attention

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Abstract—Fifth-generation (5G) networks provide connectivity to a massive number of devices and boost a plethora of applications in several different domains. However, the large adoption of connected devices increases attack surfaces and introduces several security threats that can severely damage physical objects and risk people's lives. Despite existing intrusion detection systems (IDSs), there are still several challenges to be addressed in the detection of cyber-attacks. For instance, while unsupervised IDSs are required to detect zero-day attacks, they usually present high false positive rates. Moreover, most existing IDSs rely on long short-term memory (LSTM) networks to consider time-dependencies among data. However, LSTM networks have recently been shown to present several drawbacks and limitations, which put into question their performance on sequence modeling tasks. Thus, in this paper, we investigate generative adversarial networks (GANs), a promising unsupervised approach to detecting attacks by implicitly modeling systems, and alternatives to LSTM networks to consider temporal dependencies among data. We propose a novel unsupervised GAN-based IDS that uses temporal convolutional networks (TCNs) and self-attention to detect cyber-attacks. The proposed IDS leverages edge computing and is proposed for edge servers, which bring computation resources closer to end nodes. Experiment results show that our proposed IDS can be configured to satisfy different detection rate and detection time requirements. Moreover, they show that our IDS is more accurate and at least 3.8 times faster than two state-of-the-art GAN-based IDSs that are used as baselines.

Index Terms—Intrusion detection system (IDS), Internet of Things (IoT), machine learning, generative adversarial network (GAN), temporal convolutional network (TCN), self-attention.

I. INTRODUCTION

THE INCREASING growth of connected devices, such as sensors, actuators, home appliances and vehicles is changing how we interact with our surroundings. It is reducing the gap between the physical and digital worlds and integrating devices into large-scale platforms that acquire and process data to produce automated decisions while also generating knowledge and information [1], [2]. Smart and connected devices compose smart cities, Industry 4.0, and, in general, the Internet of Things (IoT). They create a whole new world of services and applications, such as intelligent traffic lights, automated water treatment plants, and personal health monitoring applications [3], [4]. Moreover, they are expected to grow even further with the adoption of the fifth-generation of wireless/mobile communications (5G), since 5G can provide connectivity to a massive number of devices with highly diverse requirements [5], [6].

On the other hand, the broadcast nature of wireless communications enables attackers to eavesdrop on the network, inject malicious data into it, and launch cyber-attacks [7]. Therefore, the widespread adoption of IoT introduces several security threats that may impair network integrity and cause inaccurate sensing and control of systems. Such vulnerabilities could severely damage physical objects and risk people's lives [8], [9]. Despite numerous security solutions being available for the traditional Internet, the IoT's physical constraints and highly heterogeneous environment impose new security challenges. For instance, the heterogeneity brought by different access technologies, applications, and requirements significantly increases the attack surfaces and the threat from new types of attacks [10], [11], [12]. On the other hand, the limited battery and computing power of most IoT devices thwart the deployment of most cryptography- and authentication-based security mechanisms [10], [13].

To overcome these challenges, intrusion detection systems (IDSs) have emerged as a fundamental component to protect and secure networks and systems. They detect cyber-attacks when other security mechanisms fail [14], [15], [16]. In contrast to other detection approaches, anomaly-based IDSs detect

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cyber-attacks by measuring deviations between data patterns and what is considered normal behavior. Although recent advances in machine learning foment new IDS mechanisms to detect cyber-attacks [10], [17], [18], there are still several challenges to be addressed.

First, sophisticated distributed cyber-attacks, such as modern distributed denial-of-service (DDoS) attacks, significantly challenge current IDSs, as they might have multiple steps and be launched on different applications and devices. DDoS attacks attempt to exhaust a system's or network's resources by, for example, forcing multiple compromised devices to unnecessarily request resources so that there are no resources left for legitimate users. Google, Amazon Web Services, domain name system (DNS) providers, and many others have been the target of DDoS attacks. For instance, recently, a DDoS attack on a large DNS provider caused disruptions to many services, such as Airbnb, Netflix, PayPal, Visa, Amazon, The New York Times, and GitHub [19], [20]. In addition, cyber-criminals have threatened several organizations with DDoS incursions unless extortion demands are met. In 2021, such attacks disrupted Internet service providers and voice-over-IP (VoIP) operations worldwide [21], [22].

Moreover, since new attacks are constantly being launched, IDSs must be able to detect zero-day attacks, for which there is no data available. Even for known attacks, it is challenging, time-consuming, and sometimes impossible to obtain labeled data. Therefore, unsupervised IDSs, which detect both known and zero-day attacks without relying on labeled attack data, are deemed the best to use [23], [24], [25], [26]. However, most existing state-of-the-art unsupervised IDSs usually have high false positive rates and long detection times [15], [24], [27], which make them unsuitable for latency constrained applications.

Furthermore, most existing IDSs rely on long short-term memory (LSTM) networks to consider time dependencies among data, which are present in a considerable amount of real-world data, including network traffic. However, recent studies show that LSTM networks present several drawbacks, which put in doubt their status as the standard architecture for sequence modeling tasks [28], [29], [30], [31]. For instance, they process data sequentially, which significantly increases their computational complexity and challenges their performance on devices with limited computational power [32]. Moreover, LSTM networks can easily consume a lot of memory just to store the partial results of multiple cell gates during training [29]. Therefore, it is urgently necessary to propose novel unsupervised IDSs that are capable of tackling the aforementioned challenges while avoiding the drawbacks of LSTM networks.

A. Related Works

The work in [33] proposes a learning firewall that automatically updates itself with new rules to minimize false negatives and eliminate false positives. The authors of [16] propose FlowGuard, an intelligent defense mechanism that protects IoT networks against DDoS attacks. It identifies, classifies, and mitigates cyber-attacks by leveraging an edge-IoT architecture to meet the real-time requirements of

IoT applications. The work in [34] combines the Bayesian optimization-based Gaussian process (BO-GP) and the decision tree (DT) classification algorithm to detect botnet attacks on IoT devices. Similarly, the authors of [35] rely on a genetic algorithm to optimize a random forest model that detects botnet attacks based on their DNS queries. However, the approaches proposed in [16], [33], [34], and [35] follow a supervised learning approach so they cannot detect unknown attacks and require labeled attack data to detect known attacks.

Several other works propose unsupervised IDSs that leverage generative adversarial networks (GANs) to detect both known and unknown attacks without requiring labeled attack data. The works on generative adversarial networks-based anomaly detection (GAN-AD) [36] and multivariate anomaly detection with GAN (MAD-GAN) [15] propose GAN-based anomaly detection systems to find deviant behaviors resulting from cyber-attacks in cyber-physical systems (CPSs). They detect anomalies by combining GAN discrimination and reconstruction losses. However, they detect attacks by solving an optimization problem for every data pattern under evaluation, which significantly increases detection time and makes them unsuitable for low-latency constrained applications. On the other hand, the work in [27] proposes a low-latency unsupervised IDS for CPSs, called FID-GAN, that also uses GANs. It enhances MAD-GAN's architecture by training an encoder such that no optimization problem is solved at detection time and attacks are detected much faster than with MAD-GAN. However, it still presents considerable false positive rates.

Despite their interesting proposals, the works on FlowGuard [16], GAN-AD [36], MAD-GAN [15], and FID-GAN [27] rely on LSTM networks to consider time dependencies among data. LSTM networks are heavily used by existing IDS solutions, which then result in several drawbacks [29], [30], [32]. In contrast, the work on adversarially learned anomaly detection (ALAD) [37] proposes a GAN-based anomaly detection system that uses only fully-connected and convolutional neural networks. However, it is significantly worse than FID-GAN at detecting attacks [27].

To avoid LSTM's drawbacks, recent works have been proposing alternative architectures for considering time dependencies among data. The work in [29] proposes temporal convolutional networks (TCNs) by leveraging causal and dilated convolutions, and shows that TCNs can outperform LSTM networks in several sequence modeling tasks. The work in [30] proposes transformers by replacing recurrent networks for attention mechanisms in sequence transduction models. The authors of [31] propose an anomaly detection system for logs that uses transformers and show that transformers outperform LSTM networks in log sequences modeling. Finally, the work in [38] proposes an IDS that uses attention mechanisms adapted from the transformer's architecture and is more accurate than an LSTM-based model. However, it follows a supervised learning strategy and cannot detect zero-day attacks.

B. Contributions

In this paper, we propose a novel unsupervised IDS that uses a GAN to detect both known and zero-day attacks. GANs simultaneously train two competing neural networks,

namely, a generator and a discriminator. The generator learns the probabilistic distribution of a training set so that it can produce data similar to the training data. The discriminator, on the other hand, learns how to distinguish between real data and data produced by the generator. Thus, if the training set contains only normal data, the discriminator learns how to distinguish between normal data and anomalies regardless of whether they represent known or unknown attacks. Moreover, in contrast to most state-of-the-art unsupervised IDSs, which have high false positive rates and long detection times [15], [27], [36], our proposed IDS does not rely on LSTM networks. Instead, it uses TCNs and self-attention in the GAN generator and discriminator networks. TCNs and self-attention enable more computation parallelization, have a constant number of sequentially executed operations, and have been shown to yield more accurate results than LSTM networks in specific sequence modeling tasks [29], [30], [32], [39]. We conduct a comparative evaluation of different TCN and self-attention GAN architectures so that different trade-offs between detection rates and detection times are achieved and our IDS can be configured to satisfy different requirements. Furthermore, to achieve efficient service delivery with reduced end-to-end latency, our proposed system leverages edge computing by being deployed on edge servers closer to the network nodes under surveillance. In summary, the main contributions of our proposed TCN/self-attention GAN-based IDS are:

- 1) An unsupervised GAN-based IDS that is capable of detecting both known and zero-day attacks without relying on labeled attack data, which is difficult and sometimes impossible to obtain.
- 2) Experiments using TCNs and self-attention in a GAN to detect cyber-attacks with better detection results than existing GAN-based IDSs.
- 3) An evaluation of the trade-off between detection rates and detection times for different TCN and self-attention GAN architectures so that our proposed IDS can be configured to satisfy different requirements.

C. Organization

The remainder of this paper is organized as follows. Section II describes the DDoS threat scenario considered in our work. Section III presents our proposed architecture by describing the system model and the TCN and self-attention GAN architectures. Section IV explains the experiments that were conducted. In Section V, we present and discuss the results. Finally, Section VI concludes the paper and proposes possible future extensions to this work.

II. DDoS THREAT SCENARIO

While the goal of denial of service attacks is to prevent legitimate users from accessing specific network services and resources, they can achieve their goal by following different strategies: protocol exploration, network flooding, reflection amplification, and slow request/response. In our work, we consider DDoS attack types of all aforementioned strategies so that the adversaries' capabilities are described as follows.

Protocol exploration attacks rely on protocol features and implementation bugs, such as the three-way handshake mechanism of the transmission control protocol (TCP). An adversary can leverage this mechanism and send a large number of SYN messages to a server without transmitting ACK messages to acknowledge the server's responses. Thus, since the server persistently waits for the non-transmitted ACK messages, its limited buffer queue is exhausted and new connections cannot be processed. On the other hand, in network flooding attacks, the adversary sends many repetitive communication requests to fill the victim's buffer until the victim cannot accept new messages and legitimate requests are disrupted. Several network protocols may be used for flooding. For instance, the adversary can send a large number of UDP packets to random ports on the victim's host so that the victim is forced to send Internet control message protocol (ICMP) packets persistently and eventually reaches a resource-exhausted condition.

Similarly, reflection amplification attacks flood the victim by leveraging third-party servers, called reflectors, that respond to requests by transmitting large responses that significantly increase network traffic. Hence, the adversary sends many requests to reflectors by spoofing their source IP with the victim's IP so that reflectors send a large amount of traffic to the victim. Finally, slow request/response attacks exhaust a victim's resources by holding the communication channel for a long time. The adversary establishes multiple valid hypertext transfer protocol (HTTP) connections with a victim and segments legitimate HTTP packets into tiny fragments sent in each connection as slowly as possible within the maximum allowed communication time. Thus, as all victim's sockets are taken up, the victim becomes unavailable for legitimate connections.

In addition to the different strategies they can adopt, denial of service attacks become extra powerful and difficult to detect and trace back when they are launched from distributed sources with spoofed IPs. Moreover, adversaries usually take advantage of botnets, i.e., networks of computers infected by malware that can carry out commands under the attacker's control, to generate a significant amount of traffic from systems spread across the Internet. When large botnets are used, each system may only need to send out a small amount of traffic to produce enough volume to saturate the target network, making it extremely difficult to distinguish between DDoS and legitimate traffic. Therefore, DDoS attacks significantly impact network service and management while being very challenging to detect.

III. PROPOSED IDS ARCHITECTURE

In this section, we briefly explain how GANs work and how they can leverage TCNs and self-attention to consider dependencies among data. Moreover, we describe the architecture of our proposed IDS and the different configurations it can adopt to achieve different trade-offs between detection rates and detection times. Finally, we present our system's deployment architecture.

A. GAN-Based IDSs

GANs are powerful frameworks for training generator and discriminator neural networks. When trained with only normal

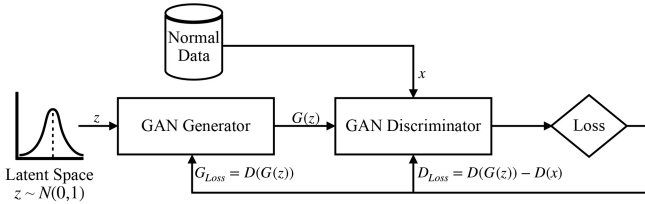


Fig. 1. The WGAN training framework used in our proposed IDS.

data, the generator implicitly models the system and learns how to produce data patterns similar to those of normal data. It learns to map random vectors z , drawn from a distribution $P(z)$ in a latent Z -space, to data patterns similar to those of normal data so that $x_{fake} = G(z)$. On the other hand, the discriminator learns to distinguish between real normal data patterns, x_{real} , and data patterns produced by the generator, x_{fake} . Thus, the discriminator's output, $D(x)$, indicates whether a data sample x is real or produced by the GAN generator, i.e., it measures deviations from normal behavior and hence detects cyber-attacks regardless of whether they are known or unknown.

The generator and discriminator neural networks are trained together in an adversarial process so that the generator tries to maximize the probability of producing outputs that are recognized as real and the discriminator tries to minimize that same probability. In our proposed system, we train a GAN according to the Wasserstein GAN (WGAN) framework, in which the generator maximizes $G_{Loss} = D(G(z))$ and the discriminator minimizes $D_{Loss} = D(G(z)) - D(x)$. In contrast to the original GAN formulation, the WGAN is easier to train and does not suffer from the gradient vanishing problem [40], [41]. Figure 1 shows the adopted WGAN's training mechanism.

Existing GAN-based IDS solutions rely heavily on LSTM networks to consider temporal dependencies among data. However, since LSTM's sequential data processing significantly increases computational complexity and memory consumption during training, recent studies have been investigating alternative approaches for sequence modeling tasks. In our work, we investigate and propose replacing LSTM networks by TCNs and self-attention in both the GAN generator and discriminator for cyber-attack detection.

B. TCNs

TCNs refer to modified convolutional architectures for sequence prediction tasks. They map input sequences to output sequences of the same length and use causal convolutions, i.e., convolutions that use only information from the past. Thus, an output at time t is convolved only with elements from times earlier than t in the previous layer. In addition, since sequence modeling tasks may require more history, TCNs also use dilated convolutions to enable the architecture to look far into the past. Thus, for an input sequence $x \in \mathbb{R}^n$ and a filter $f : \{0, \dots, k-1\} \rightarrow \mathbb{R}$, the dilated convolution on element s of the sequence is defined as

$$F(s) = (x *_{d} f)(s) = \sum_{i=0}^{k-1} f(i)x_{s-di}, \quad (1)$$

where k is the filter size, d is the dilation factor, and $*_{d}$ is the dilated convolution operation. The dilation factor indicates how far into the past convolution operation $*_{d}$ looks. Hence, while $d = 1$ reduces Equation (1) to a regular convolution operation, the larger the dilation factor, the further back $*_{d}$ looks. Finally, TCN networks allow a residual connection so the architecture learns what modifications are imposed on the data rather than only modifying it. This connection contributes to avoiding the gradient vanishing problem and consists of adding the input x to the output of a series of transformations T . It is given by

$$O(x) = \Phi(x + T(x)), \quad (2)$$

where Φ is an activation function.

TCNs provide a powerful way to extract temporal dependencies from data and have been shown to have several advantages over LSTM networks for modeling sequences. More specifically, when our proposed solution uses TCNs, the filter f convolves across a sequence of incoming network flows by considering features only from network flows that have already occurred (causal convolution). A dilation factor is also considered (dilated convolution) so that network flows that occurred a long time ago are also taken into account. Computations can be performed in parallel since the same filter can be used in all layers, and input sequences can be processed as a whole. This means TCNs do not need to store the partial results of computations and thus consume less memory during training. Finally, TCNs have been shown to have stabler gradients, which avoids the gradient vanishing and exploding problems [29], [32].

C. Self-Attention

Attention functions are defined as the mapping of a matrix of queries Q , a matrix of keys K , and a matrix of values V to an output. Scaled dot product attention is one type of attention function, which computes a context matrix C as

$$C = \text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (3)$$

where d_k is the dimension of values. Matrices K and V usually correspond to input sequences x , whereas matrix Q is composed of randomly initialized trainable parameters. The dot product of Q and K^T gives a measure of the pairwise similarity between the query and key matrices, which results in an attention score. Thus, the C matrix represents the intrinsic dependencies between representations of a sequence.

Moreover, it has been shown that using linearly projected queries, keys, and values h times with learned linear projections contributes to extracting relationships between data [30], [39]. Thus, multi-head attention (MHA) modules perform attention functions in parallel on each of the projected versions of queries, keys, and values, and then concatenate their outputs as

$$\text{MHA}(Q, K, V) = W^0 \text{Concat}(\text{head}_1, \dots, \text{head}_h), \quad (4)$$

where W^0 is a parameter matrix for the concatenation operation and $\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$. W_i^Q , W_i^K , and W_i^V are parameter matrices that project queries,

TABLE I
COMPUTATIONAL COMPLEXITY

Layer Type	Complexity per Layer	Sequential Operations
LSTM	$O(nd^2)$	$O(n)$
TCN	$O(knd^2)$	$O(1)$
Self-Attention	$O(n^2d)$	$O(1)$
MHA	$O(n^2d + nd^2)$	$O(1)$

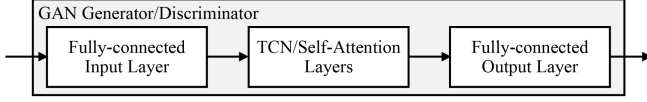


Fig. 2. The GAN generator and discriminator architectures.

keys, and values, respectively. Finally, self-attention considers that all the keys, values, and queries come from the same place, such as the output of the previous layer in a neural network. This allows modules to capture in-depth contextual information and relationships between data.

Similarly to TCNs, attention mechanisms make it possible to extract dependencies among data and have been shown to outperform LSTM networks in several sequence modeling tasks. They are more capable of extracting features than LSTM networks, which contributes to more accurate models [39]. In addition, they can process sequences as a whole and they enable more computation parallelization as MHA heads can run in parallel. Furthermore, while LSTM networks require $O(n)$ sequential operations, TCN, self-attention, and MHA layers have a constant number of sequentially executed operations. Table I summarizes the computational complexity of LSTM, TCN, self-attention, and MHA layers, where n is the sequence length, d is its depth, and k is the kernel size of convolutions [30], [42].

D. Proposed Detection Architecture

Our proposed architecture consists of a GAN that relies on TCNs and self-attention to consider time dependencies among data. Since different applications may have different requirements and constraints, we propose different architectures for the GAN generator and discriminator neural networks so that different trade-offs are achieved between detection rates and detection times. More specifically, we design generator and discriminator networks with one fully connected input layer, one fully connected output layer, and hidden layers of one or more TCN or self-attention blocks. Figure 2 shows the proposed high-level architectures of the GAN generator and discriminator.

The proposed TCN block allows the architecture to learn from experience and consists of a single dilated causal convolution and a rectified linear unit (ReLU) activation function. In addition, to avoid overfitting, it has a normalization layer and a dropout layer for regularization. This block can be replicated N times such that a single convolution layer is responsible for the TCN residual connection. The number of dilated causal convolutions, i.e., the value of N , directly impacts the detection rates and detection times. While higher values of N may increase our IDS's ability to learn and detect attacks, it also increases detection times, as the more layers there are, the

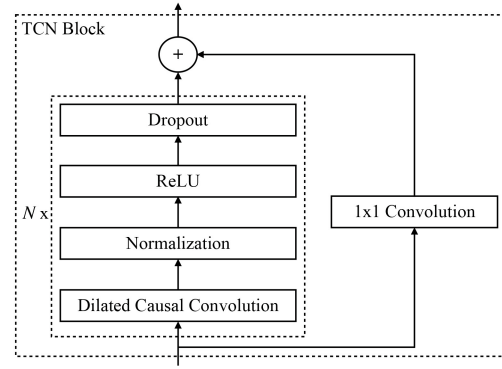


Fig. 3. The TCN block architecture.

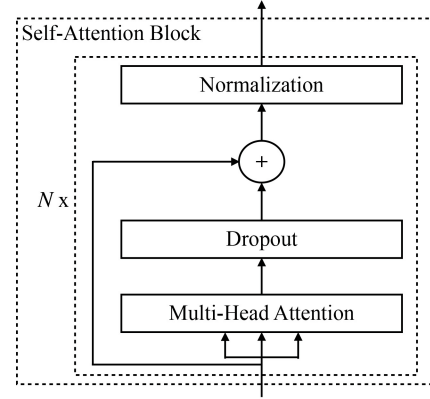


Fig. 4. The self-attention block architecture.

longer the detection times. Figure 3 shows the TCN block architecture.

On the other hand, the proposed self-attention block consists of an MHA module that uses self-attention. Similarly to the TCN block, normalization and dropout layers are used to avoid overfitting. Moreover, a residual connection is included to help with the network's training, as it allows gradients to flow through the network. Finally, N self-attention blocks can be cascaded to increase our IDS's ability to learn and detect attacks, at the expense of also increasing detection time. Figure 4 shows the self-attention block architecture.

E. Proposed Deployment Architecture

Edge computing provides cloud computing capabilities closer to consumers and the data generated by applications. It is one of the main pillars for meeting low latency and bandwidth efficiency requirements [43], [44]. More specifically, edge computing architectures introduce edge servers, to which devices can offload computational tasks and receive back their results in real time [16], [45]. Thus, our proposed IDS is deployed as an application on edge servers so that IoT devices can send their network flows for analysis and receive back attack detection alerts. Depending on their criticality and sensitivity, IoT devices may be configured to adopt different actions upon receiving alerts, such as dropping packets, resetting connections, or blocking the traffic from suspicious nodes.

We suggest using the open-source Kafka-ML [46] framework to deploy our IDS's machine learning models and transfer data between IoT devices and edge nodes. It uses data streams to manage machine learning pipelines and the Apache Kafka distributed publish/subscribe messaging system [47] to transfer large amounts of data with low latency. In addition, using the framework in all edge nodes and the cloud significantly reduces the machine learning models' response time [48], [49]. Finally, Kafka-ML relies on containerization and container orchestration platforms to ensure portability, easy distribution, and high availability.

Moreover, we propose that instances of our IDS that are deployed on edge servers in different regions interact with each other to exchange attack detection reports so that they become aware of attacks occurring in neighboring regions. In addition, they also send attack detection reports to a cloud service, which has a large-scale view and becomes aware of attacks that simultaneously target different locations. This awareness is essential for evaluating the potential risks and impacts of sophisticated distributed cyber-attacks to take appropriate countermeasures. Figure 9 in Appendix A shows the deployment of our proposed IDS on multiple edge servers.

IV. METHODOLOGY AND PERFORMANCE EVALUATION

In this section, we briefly present the dataset used in our experiments, which contains both normal and attack data. Then, we explain the models implemented and experiments conducted.

A. Dataset Presentation

In order to evaluate our proposed IDS, we use the CICDDoS2019 dataset provided by the Canadian Institute for Cybersecurity (CIC) and the University of New Brunswick (UNB) [50], [51]. This dataset contains benign traffic data and the most modern and common DDoS attacks, such as Syn, UDP, UDPLag, MSSQL, NetBIOS, LDAP, and Portmap, covering the four DDoS attack strategies presented in Section II. The dataset provides 83 network flow features extracted from raw traffic data using the CICFlowMeter-V3 tool [52]. While many traditional network-based IDSs rely on deep packet inspection, this approach is computationally costly and challenging to implement when network traffic is encrypted [53]. Thus, like most state-of-the-art IDSs [16], [27], [54], in our work, we rely on network flow features to detect malicious activities. We use the 35 most relevant network flow features from those defined in [16], such as flow duration and the total number of packets in the forward and backward directions, as well as five features that identify network flows: source IP, destination IP, source port, destination port, and protocol. Table IX in Appendix B lists all the features used in our work.

To train and evaluate our IDS, we constructed a training, a validation, and a testing set. The training set, which is used to train the GAN, was constructed by sampling 80% of the normal network flows of a training day defined by the dataset. The validation set, which is used to optimize our models' hyper-parameters, was formed by the remaining 20% of the normal network flows of the training day in question and

TABLE II
TRAINING, VALIDATION, AND TESTING SETS

	Normal Network Flows	DDoS Attacks Network Flows
Training Set	45,408	0
Validation Set	11,342	68,052
Testing Set	50,000	50,000

TABLE III
MALICIOUS NETWORK FLOWS PER DDoS ATTACK TYPE

DDoS Attack Type	Training Set	Validation Set	Testing Set
Syn	0	11,342	8,021
UDP	0	11,342	8,021
UDPLag	0	11,342	1,873
MSSQL	0	11,342	8,021
NetBIOS	0	11,342	8,021
LDAP	0	11,342	8,021
Portmap	0	0	8,022

DDoS attacks sampled from the training day. Finally, the testing set, which is used to evaluate our IDS's performance, was constructed by sampling 50,000 normal network flows and 50,000 malicious network flows from a testing day defined by the dataset. Since our testing set contains samples collected from a different day and is only used after the training of all models has been completed, it provides unbiased results. Moreover, although the testing set's malicious network flows result from various types of DDoS attacks, our IDS relies on a binary classifier that distinguishes between normal and malicious network flows rather than classifying by attack type. Hence, the testing set is considered balanced. Table II depicts the number of normal and DDoS network flows in the constructed sets. Table III indicates the number of malicious network flows per DDoS attack type.

As shown in Tables II and III, our proposed IDS relies only on normal network flows to train its neural networks. Although malicious network flows of the Syn, UDP, UDPLag, MSSQL, NetBIOS, and LDAP attacks are present in the validation set, they are used only to tune the models' hyper-parameters. Moreover, since the Portmap type of DDoS attack is present only in the testing set, it represents a zero-day attack for which no information is available.

B. Simulation Experiments

We conducted multiple experiments by training and evaluating the GAN depicted in Figure 2 using different numbers of TCN and self-attention blocks, which are depicted in Figures 3 and 4, respectively. In addition, we trained the GAN using LSTM networks instead of the proposed blocks as hidden layers to have a baseline for comparing our IDS's performance. All models were optimized using the Optuna framework [55], which automatically searches for optimal hyper-parameter values by trial and error, and employed the early stopping mechanism to avoid overfitting. Several hyper-parameters were tuned, such as learning rates, optimizer, batch size, hidden dimension, kernel of convolutions, and number of attention heads. Moreover, we experimented with concatenating several layers of the TCN and self-attention blocks by varying the parameter N defined in Figures 3 and 4. Finally,

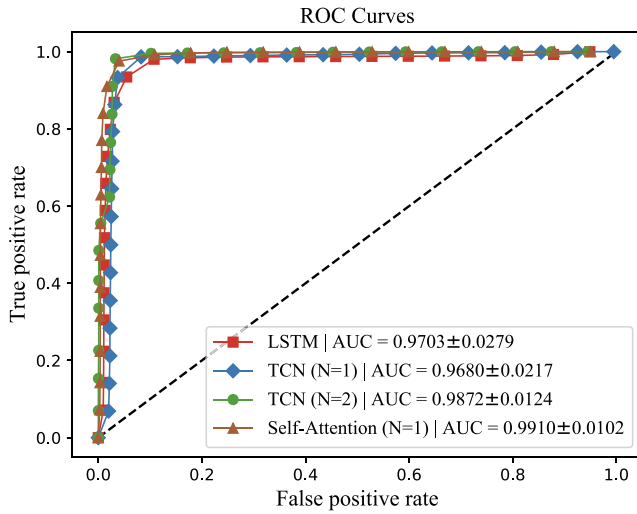


Fig. 5. Our proposed IDS's ROC curves.

we replicated our training experiments twenty times to reduce bias from the stochastic training. Table X in Appendix C lists the hyper-parameter values used in our work.

V. RESULTS AND DISCUSSION

In our experiments, we evaluated the detection rate, detection time, and complexity of our proposed IDS when using one or more TCN and self-attention blocks as hidden layers. The goal was to identify a trade-off between detection rates and detection times so that our IDS can employ different configurations and satisfy different requirements. In addition, we evaluated whether TCN and self-attention blocks outperform LSTM networks in our proposed GAN-based IDS. Finally, we compared our IDS to two state-of-the-art GAN-based IDSs: FID-GAN [27] and ALAD [37]. FID-GAN considers temporal dependencies among data by using LSTM networks in both the GAN generator and discriminator. ALAD, on the other hand, does not use LSTM networks or consider time dependencies among data. It relies only on fully connected and regular convolutional networks.

A. Detection Rates

We used the area under the curve (AUC) of the receiver operating characteristic (ROC) curve (collectively, AUCROC) as the metric to evaluate our proposed IDS's cyber-attack detection performance on the testing set samples. Each point on the curves represents both the true positive and false positive rates achieved for a threshold. Hence, the AUCROC metric allows us to evaluate our solution for many different thresholds at once rather than for only one at a time. Moreover, it shows which threshold yields the best results in terms of maximizing the true positive rate or minimizing the false positive rate.

Figure 5 shows the ROC curves obtained when using LSTM networks, one TCN block, two TCN blocks, and one self-attention block as the hidden layers in our proposed architecture. Other TCN and self-attention block configurations did not improve the AUCROC results but increased the detection

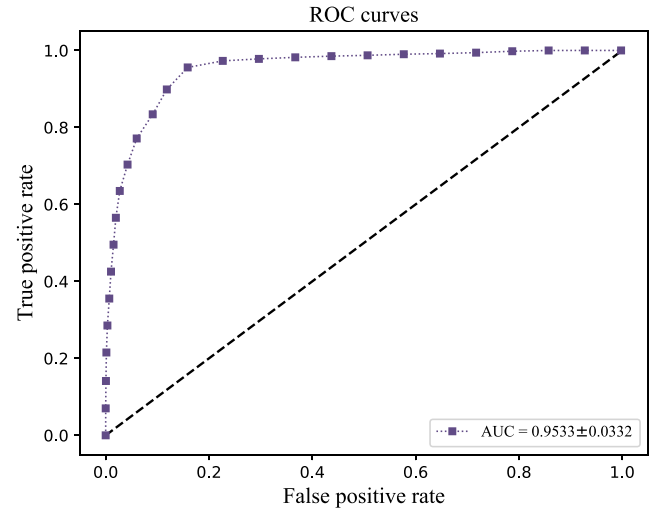


Fig. 6. ALAD's ROC curve.

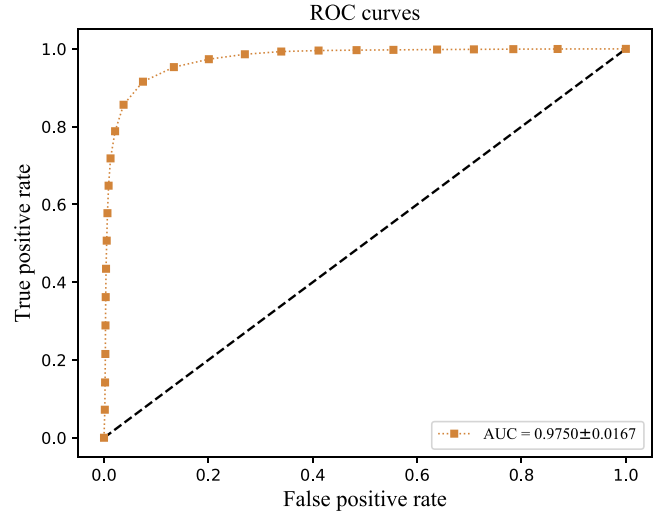


Fig. 7. FID-GAN's ROC curve.

times, as the IDS takes longer to detect attacks the more layers it has. The plots verify that our IDS achieves the highest AUCROC results when using two TCN blocks or one self-attention block. Moreover, since the AUCROC values achieved are close to 1, our IDS ensures low false positive and false negative rates simultaneously. While minimizing false positives is essential for keeping the network operational, minimizing false negatives is essential for ensuring security.

Moreover, we compared the AUCROC results of our proposed IDS to those of ALAD [37] and FID-GAN [27]. Our proposed IDS outperforms ALAD in all its configurations, as ALAD uses only fully connected and regular convolutional layers, and does not consider dependencies among data. It also outperforms FID-GAN, which relies on LSTM networks, when it is configured with two TCN blocks or a single self-attention block. ALAD's and FID-GAN's ROC curves are shown in Figures 6 and 7, respectively.

Furthermore, we also conducted hypothesis tests to verify whether there were statistical differences between the AUCROC of our proposed IDS, ALAD, and FID-GAN. Hence, we first conducted D'Agostino and Pearson's hypothesis test

TABLE IV
TUKEY'S HSD PAIRWISE GROUP COMPARISONS
(95.0% CONFIDENCE INTERVAL)

Comparison		p-value
Self-Attention	TCN (N=2)	0.989
Self-Attention	TCN (N=1)	0.017
Self-Attention	LSTM	0.011
Self-Attention	FID-GAN	0.212
Self-Attention	ALAD	0.000
TCN (N=2)	TCN (N=1)	0.024
TCN (N=2)	LSTM	0.012
TCN (N=2)	FID-GAN	0.349
TCN (N=2)	ALAD	0.000
TCN (N=1)	LSTM	0.999
TCN (N=1)	FID-GAN	0.920
TCN (N=1)	ALAD	0.299
LSTM	FID-GAN	0.973
LSTM	ALAD	0.065
FID-GAN	ALAD	0.029

to verify whether a normal distribution could approximate the AUCROC values obtained for each model. This verification allowed us to conduct the one-way ANOVA hypothesis test to verify whether there was a significant difference between at least two of the models evaluated. The ANOVA test confirmed that at least one of the models differed from the others, such that there was a statistically significant difference among them. Since ANOVA cannot determine which model differed from the others, we conducted Tukey's honestly significant difference (HSD) post hoc test, which evaluates the models two-by-two. Table IV shows the p-values of the models' comparison obtained from the post hoc test. We reject the null hypothesis that there is no significant difference between the model's AUCROCs whenever the post hoc test p-value does not exceed 0.05. Thus, our results show that when it uses self-attention or two TCN layers, our IDS is statistically different from its LSTM-based version. Therefore, self-attention and two TCN blocks successfully replace LSTM networks for attack detection and achieve better detection results.

Finally, we evaluated our proposed IDS, ALAD, and FID-GAN at the equal error rate (EER), which corresponds to the threshold at which the false positive and false negative rates are equal. Table V shows the accuracy, precision, recall, and F1-scores of our IDS in four different configurations, ALAD, and FID-GAN. While our IDS outperforms ALAD and FID-GAN in all its configurations according to all the metrics used, it achieves the best results when it is configured with two TCN blocks. Finally, although our goal is not to identify different types of attacks, we provide in Table VI the overall normal and attack (recall) detection rates as well as the detection rates for each type of DDoS attack in the testing set. Our IDS can detect the Portmap attack, which represents a zero-day attack, with a detection rate as high as 0.9993, which is higher than it can achieve for the other attack types. Therefore, our proposed IDS is considered able to detect unknown attacks.

B. Detection Times

To evaluate how long our IDS takes to detect attacks, we measured its mean detection time when using LSTM networks, TCN blocks, and self-attention blocks. The configuration with

TABLE V
ACCURACY, PRECISION, RECALL, AND F1-SCORES
OF OUR IDS, ALAD, AND FID-GAN

	Accuracy	Precision	Recall	F-1
LSTM	0.9405	0.9405	0.9405	0.9405
TCN Block (N=1)	0.9588	0.9588	0.9588	0.9588
TCN Block (N=2)	0.9707	0.9705	0.9710	0.9707
Self-Attention Block (N=1)	0.9682	0.9682	0.9682	0.9682
FID-GAN	0.9203	0.9203	0.9203	0.9203
ALAD	0.8860	0.8860	0.8860	0.8860

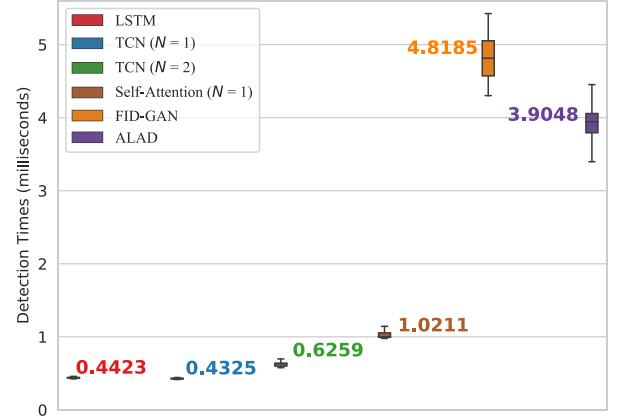


Fig. 8. Detection times of our IDS, ALAD, and FID-GAN.

a single TCN block had the shortest detection time, hence it is the preferred configuration for latency constrained applications. Moreover, we compared the detection times our solution achieved to those of ALAD and FID-GAN, which took longer than our proposed solution to detect attacks. Particularly, FID-GAN has a much longer detection time than our IDS because it relies on a more complex GAN architecture and computes a reconstruction loss using an encoder neural network. Similarly, such complexity and the need for training an encoder neural network in addition to the GAN make the training time of FID-GAN much longer than that of our IDS. Therefore, our IDS is considered the best IDS of the three. Figure 8 shows the detection times of our proposed IDS, ALAD, and FID-GAN.

Furthermore, the results in Figures 5 and 8 verify a trade-off between detection rates and detection times. For instance, our IDS achieves the highest AUCROC value and the longest detection time when configured with a single self-attention block. On the other hand, it achieves the lowest AUCROC value and the shortest detection time when configured with a single TCN block. Therefore, depending on the application's requirements and whether it is more important to achieve higher detection rates or shorter detection times, our IDS can be configured with different blocks as hidden layers and satisfy different constraints.

C. Complexity Analysis

To evaluate the complexity of our IDS, we present in Table VII the number of epochs trained, the mean training time per epoch, the total convergence time, and the number of parameters in the GAN discriminator and generator. In addition, we detail in Table VIII the number of parameters of each

TABLE VI
DETECTION RATES BY DDoS ATTACK TYPE

	Normal	Attack	Syn	UDP	UDPLag	MSSQL	NetBIOS	LDAP	Portmap
LSTM	0.9405	0.9405	0.7290	0.9994	0.9215	0.9999	0.9479	0.9996	0.9718
TCN Block (N=1)	0.9588	0.9588	0.9728	0.9946	0.6610	0.9868	0.935544	0.9994	0.9333
TCN Block (N=2)	0.9704	0.9710	0.8242	1.0000	0.9856	1.000	0.9990	1.000	0.9993
Self-Attention Block (N=1)	0.9682	0.9682	0.8897	1.0000	0.9770	1.000	0.9439	0.9998	0.9739
FID-GAN	0.9203	0.9203	0.6052	0.9516	0.8655	0.9898	0.9925	0.9998	0.9960
ALAD	0.8860	0.8860	0.5726	0.8193	0.8831	0.9662	0.9736	0.9999	0.9848

TABLE VII
COMPUTATIONAL COMPLEXITY OF EACH CONFIGURATION OF OUR IDS

	LSTM	TCN (N=1)	TCN (N=2)	Self-Attention (N=1)
Number of Epochs	33	50	45	42
Training Time (s/epoch)	12.03	10.07	13.64	19.42
Convergence Time (s)	397.00	503.29	613.80	815.80
Number of Parameters on Discriminator	4,234	3,101	4,961	5,921
Number of Parameters on Generator	3,618	2,330	8,180	5,520

TABLE VIII
NUMBER OF PARAMETERS OF EACH CONFIGURATION OF OUR IDS

		LSTM	TCN (N=1)	TCN (N=2)	Self-Attention (N=1)
Number of Parameters on Discriminator	Input Layer	1,640	1,640	1,640	1,640
	Hidden Layers	2,568	1,450	3,300	3,440
	Output Layer	26	11	21	841
Number of Parameters on Generator	Input Layer	840	440	4,040	440
	Hidden Layers	2,568	1,450	3,300	3,440
	Output Layer	210	440	840	1,640

model's input layer, hidden layers, and output layer. The models are trained for different numbers of epochs due to the early stopping mechanism, the LSTM model trains for the fewest number of epochs and converges fastest. However, the model with a single TCN block has the lowest number of parameters and the shortest training time per epoch, which is reflected in the detection time results in Figure 8, as it has the shortest detection times. Finally, even though the self-attention model converges slowest, it achieves the best detection rates.

Usually, the more parameters a neural network has, the longer it takes to converge. However, although a few works have proposed a correlation between the number of parameters of a neural network and its convergence time, they are not too accurate and depend on many other variables, such as the optimizer and the number and complexity of training samples [56], [57]. For that reason, dynamic learning techniques and the early stopping mechanism have been developed and largely adopted to let the models decide when they have learned enough and must stop training to reduce the training time and avoid overfitting [58], [59]. Therefore, although all models evaluated were given the same computational resources and training time, so they had the same computational budget, they were allowed to stop training earlier, i.e., after different numbers of epochs using the early stopping mechanism as it is commonly done in machine learning. Precisely, all the models had a maximum time of 15 minutes (900 seconds) to train on an AMD Ryzen Threadripper 1920X 12-core processor 2.2GHz with 64GB of RAM and an NVIDIA GeForce RTX 2080 in a Pytorch environment. On the other hand, providing less time than what is required for training the models, i.e., forcing them to stop training before convergence, would unnecessarily compromise their detection results as we adopt an offline training procedure that deploys models only after they have been trained. Moreover, even though we consider

15 minutes a very reasonable maximum training time, since the models will only work and detect malicious network flows after they have been trained, our primary concern is not the training time but the detection time.

D. Combining Protection Techniques

Despite our solution's results, security is usually constructed in layers to enhance protection against cyber-attacks. Thus, our proposed IDS may be combined with other techniques to protect against DDoS attacks. For instance, we can limit attack surface areas by not exposing applications and resources to ports and protocols from which they do not expect to receive any communication [60]. Moreover, we can rely on firewalls, Web application firewalls (WAF), and traditional signature-based IDSs, which are rule-based, to reduce the burden on our proposed IDS [61]. Finally, we can rely on scalable architectures that quickly adjust their resources to accommodate high traffic volumes and maintain availability in critical systems [60].

Furthermore, DDoS attacks are usually launched from botnets to create large volumes of traffic. Therefore, a fundamental aspect of protecting against DDoS attacks is thwarting botnet recruitment, which requires protection techniques, such as limiting attack surfaces, and using firewalls and botnet detection systems [62], [63]. In future works, we will combine our proposed IDS with botnet detection techniques.

E. Strengths and Limitations

One of the main strengths of our proposed IDS is its ability to be more accurate and at least 3.8 times faster than the two state-of-the-art GAN-based IDSs we used as baselines. Moreover, it can use different hidden layers to satisfy different requirements depending on whether it is more important to have higher detection rates or shorter detection times. Finally,

our solution follows an unsupervised approach so that it does not require labeled attack data and can detect unknown attacks, such as Portmap attacks.

On the other hand, our IDS is limited to detecting attacks, as mitigating them is outside of the scope of our work. In addition, it has not been combined with other protection mechanisms, such as botnet detection techniques, which we will investigate in future works. Finally, our IDS uses only the discriminator's output. However, the reconstruction loss, which is computed using the GAN generator, could improve detection rates at the expense of increasing the detection time as is noted in [15], [27].

Furthermore, another limitation of our proposed IDS is its offline training procedure. Since the normal behavior of systems and networks under surveillance may change with time, our IDS needs to be retrained from time to time so that it keeps up to date. However, since large environments usually have a massive amount of network data, which may reach several gigabytes per hour, constantly retraining our IDS might be operationally challenging. In addition, acquiring training data from multiple nodes may raise privacy issues as such data may contain sensitive information that must not be shared. In the face of those limitations, in future works, we will investigate and propose an online federated training procedure for our IDS that leverages federated learning to preserve privacy while always being up to date.

VI. CONCLUSION

In this paper, we propose a novel unsupervised GAN-based IDS that is capable of detecting both known and zero-day attacks without relying on labeled attack data. In contrast to most existing IDSs, which rely on LSTM networks, our proposed architecture considers dependencies among data by relying on TCNs and self-attention. In our experiments, we verify the trade-off between detection rates and detection times for different configurations of our IDS. Our solution can be configured to satisfy different requirements depending on whether it is more important to achieve higher accuracies or shorter detection times. Moreover, our simulation experiments show that our proposed IDS achieves higher AUCROC values and shorter detection times than two state-of-the-art GAN-based IDSs. Therefore, not only does our IDS achieve better detection rates than LSTM-based IDSs, it is also more suitable than them for latency constrained applications.

Finally, although variational autoencoders (VAEs) are conceptually different than GANs, they have also yielded promising results in terms of learning data representations and detecting malicious activities [64], [65]. Thus, in future works, we will investigate the use of VAEs and combinations of VAEs and GANs for unsupervised attack detection, and combine them with botnet detection techniques. Moreover, we will propose an online federated training procedure so that our IDS is constantly retrained and kept up to date while preserving privacy by sharing only the weights of neural networks in different nodes instead of sensitive data. Furthermore, we will construct a new DDoS dataset with more DDoS attack types compared to the existing datasets to better evaluate

TABLE IX
FEATURES USED

Feature	Description
Source IP	Flow source IP
Source Port	Flow source port
Destination IP	Flow destination IP
Destination Port	Flow destination port
Protocol	Flow protocol
Flow duration	Flow duration in microseconds
Total Fwd Packet	Total packets in the forward direction
Total Bwd packets	Total packets in the backward direction
Total Length of Fwd Packet	Total size of packet in the forward direction
Total Length of Bwd Packet	Total size of packet in the backward direction
Fwd Packet Length Max	Maximum size of packet in the forward direction
Fwd Packet Length Min	Minimum size of packet in the forward direction
Fwd Packet Length Mean	Mean size of packet in the forward direction
Fwd Packet Length Std	Standard deviation size of packet in the forward direction
Bwd Packet Length Max	Maximum size of packet in the backward direction
Bwd Packet Length Min	Minimum size of packet in the backward direction
Bwd Packet Length Mean	Mean size of packet in the backward direction
Bwd Packet Length Std	Standard deviation size of packet in the backward direction
Flow Byte/s	Number of flow bytes per second
Flow Packets/s	Number of flow packets per second
Fwd Packets/s	Number of forward packets per second
Bwd Packets/s	Number of backward packets per second
Flow IAT Max	Maximum time between two packets sent in the flow
Flow IAT Min	Minimum time between two packets sent in the flow
Flow IAT Mean	Mean time between two packets sent in the flow
Flow IAT Std	Standard deviation time between two packets sent in the flow
Fwd IAT Max	Maximum time between two packets sent in the forward direction
Fwd IAT Min	Minimum time between two packets sent in the forward direction
Fwd IAT Mean	Mean time between two packets sent in the forward direction
Fwd IAT Std	Standard deviation time between two packets sent in the forward direction
Fwd IAT Total	Total time between two packets sent in the forward direction
Bwd IAT Max	Maximum time between two packets sent in the backward direction
Bwd IAT Min	Minimum time between two packets sent in the backward direction
Bwd IAT Mean	Mean time between two packets sent in the backward direction
Bwd IAT Std	Standard deviation time between two packets sent in the backward direction
Bwd IAT Total	Total time between two packets sent in the backward direction
Fwd PSH flag	Number of times the PSH flag was set in packets travelling in the forward direction (0 for UDP)
Bwd PSH Flag	Number of times the PSH flag was set in packets travelling in the backward direction (0 for UDP)
Fwd Header Length	Total bytes used for headers in the forward direction
Bwd Header Length	Total bytes used for headers in the backward direction

our IDS's generalization and performance. For instance, we will consider the detection of DoS attacks caused by adversarial attacks that compromise the functionality of machine learning models, such as the one proposed in [66], which interrupts wireless communications by compromising machine learning-based modulation classifiers on wireless receivers.

APPENDIX A DEPLOYMENT ARCHITECTURE

Figure 9 shows the deployment of our proposed IDS on multiple edge servers.

APPENDIX B FEATURES USED

Table IX lists the features used in our work.

TABLE X
HYPER-PARAMETER VALUES

	LSTM	TCN (N=1)	TCN (N=2)	Self-Attention (N=1)
Maximum number of epochs	50	50	50	50
Early Stopping Patience	15	15	15	15
Optimizer	Adam	Adam	Adam	Adam
Discriminator's Learning Rate	0.00284252	0.00485687	0.00753606	0.01156386
Generator's Learning Rate	0.00004508	0.00015387	0.00000151	0.00003940
Dropout	0.25	0.25	0.25	0.20
Batch Size	64	64	64	64
Latent Dimension	20	10	100	10
LSTM Hidden Dimension	20	-	-	-
TCN Number of Levels	-	1	1	-
TCN Kernel Size	-	2	2	-
Attention Heads	-	-	-	40
N	-	1	2	1

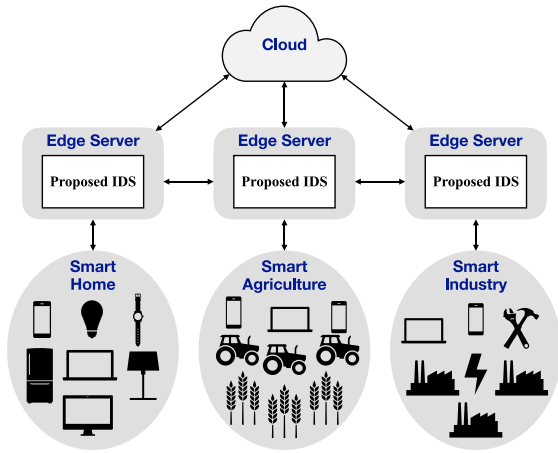


Fig. 9. Proposed deployment architecture.

APPENDIX C HYPER-PARAMETER VALUES

Table X lists the hyper-parameter values used in our work.

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